

441. Let the original duration of the tour be n days.

$$\begin{aligned}\text{Then, } \frac{360}{n} - \frac{360}{n+4} &= 3 \Leftrightarrow 360 \left[\frac{n+4-n}{n(n+4)} \right] = 3 \\ &\Leftrightarrow n(n+4) = 120 \times 4 = 480 \\ &\Leftrightarrow n^2 + 4n - 480 = 0 \Leftrightarrow (n+24)(n-20) = 0 \Leftrightarrow n = 20.\end{aligned}$$

442. Let the original number of students be x .

$$\begin{aligned}\text{Then, } \frac{500}{x-5} - \frac{500}{x} &= 5 \Leftrightarrow \frac{1}{x-5} - \frac{1}{x} = \frac{1}{100} \Leftrightarrow x(x-5) = 500 \\ &\Leftrightarrow x^2 - 5x - 500 = 0 \\ &\Leftrightarrow (x-25)(x+20) = 0 \Leftrightarrow x = 25.\end{aligned}$$

Hence, number of students who attended the picnic = $(25 - 5) = 20$.

443. Let the original cost per student be ₹ x .

$$\begin{aligned}\text{Then, } \frac{720}{x} - \frac{720}{x+9} &= 4 \Leftrightarrow 720 \left[\frac{x+9-x}{x(x+9)} \right] = 4 \\ &\Leftrightarrow x(x+9) = 1620 \\ &\Leftrightarrow x^2 + 9x - 1620 = 0 \\ &\Leftrightarrow (x+45)(x-36) = 0 \Leftrightarrow x = 36.\end{aligned}$$

444. Let the total price of the CDs be ₹ x and the number of boys be n .

$$\begin{aligned}\text{Then, } \frac{x}{n-2} - \frac{x}{n} &= 1 \Leftrightarrow \frac{nx - x(n-2)}{n(n-2)} = 1 \\ &\Leftrightarrow 2x = n^2 - 2n \Leftrightarrow x = \frac{n^2 - 2n}{2}.\end{aligned}$$

Since x is a whole number, so n must be even.

For $n = 20$, we have: $x = 180 < 200$.

For $n = 22$, we have: $x = 220$.

For $n = 24$, we have: $x = 264 > 250$.

So, $n = 22$, $x = ₹ 220$.

445. $X + 3X \times 0.50 + 14 \times 0.10 + 4X \times 0.05 = 50$
 $\therefore X + 1.5X + 1.40 + 0.2X = 50 \Leftrightarrow 2.7X = 48.60$
 $\Leftrightarrow X = 18$.

446. Let the number of 20-paise coins be x . Then, number of 25-paise coins = $(324 - x)$.

$$\begin{aligned}\therefore 0.20 \times x + 0.25(324 - x) &= 71 \\ \Leftrightarrow 20x + 25(324 - x) &= 7100 \\ \Leftrightarrow 5x = 1000 \Leftrightarrow x &= 200.\end{aligned}$$

Hence, number of 25-paise coins = $(324 - x) = 124$.

447. Let number of notes of each denomination be x .
 Then, $x + 5x + 10x = 480 \Leftrightarrow 16x = 480 \Leftrightarrow x = 30$.
 Hence, total number of notes = $3x = 90$.

448. Let the number of coins of each denomination be x .

$$\begin{aligned}\text{Then, } x + 0.5x + 0.25x + 0.10x + 0.05x &= 380 \\ \Rightarrow 1.9x = 380 \Rightarrow x &= \frac{380}{1.9} = 200.\end{aligned}$$

Hence, number of one-rupee coins = 200.

449. Let the number of 50 p-coins and 25 p-coins be x and y respectively.

Then, number of 1 rupee coins = $175 - (x + y)$.

$$\therefore 0.5x = 0.25y \text{ or } 2x = y.$$

So, number of 50 p-coins = x , number of 25 p-coins = $2x$, number of 1 rupee coins = $(175 - 3x)$.

$$\therefore 175 - 3x = 0.5x \Leftrightarrow 3.5x = 175 \Leftrightarrow x = \frac{175}{3.5} = 50.$$

$$\begin{aligned}\text{Hence, total amount} &= (175 - 3x) + 0.5x + 0.25 \times 2x \\ &= 175 - 3x + 0.5x + 0.5x = 175 - 3x + x \\ &= 175 - 2x = ₹ (175 - 2 \times 50) = ₹ 75.\end{aligned}$$

450. Total value of one 2-rupee coin, two 50-p coins and two 25-p coins

$$= ₹ (2 + 2 \times 0.50 + 2 \times 0.25) = ₹ 3.50.$$

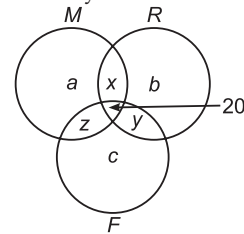
$$\text{Remaining amount} = ₹ (25 - 3.50) = ₹ 21.50.$$

For the minimum number of coins, we must have maximum amount in denominations of 2-rupee and 50-p which shall be ten 2-rupee and three 50-p coins i.e., 13 more coins.

$$\therefore \text{Required number of coins} = (1 + 2 + 2) + 13 = 18.$$

Questions 451 and 452

Let the circles M , R and F represent the number of children who took ride on merry-go-round, roller coaster and Ferris wheel only.



Let a = number of children who took ride on merry-go-round only;

b = number of children who took ride on roller coaster only;

c = number of children who took ride on Ferris wheel only;

x = number of children who took ride on both merry-go-round and roller coaster;

y = number of children who took ride on both roller coaster and Ferris wheel;

z = number of children who took ride on both merry-go-round and Ferris wheel.

Then, number of children who took exactly one ride = $a + b + c$;

number of children who took exactly two rides

$$= x + y + z;$$

number of children who took all three rides = 20;

number of children who took at least 2 rides

$$= x + y + z + 20.$$

$$\therefore \text{We have: } x + y + z + 20 = 55 \Leftrightarrow x + y + z = 35$$

$$\text{Total receipt} = ₹ [a + b + c + 2(x + y + z) + 3 \times 20]$$

$$= ₹ (a + b + c + 2 \times 35 + 60)$$

$$= ₹ (a + b + c + 130).$$

$$\therefore a + b + c + 130 = 145 \text{ or } a + b + c = 15.$$

451. Number of children who took exactly one ride

$$= a + b + c = 15.$$

452. Number of children who did not try any ride

$$= 85 - (a + b + c + x + y + z + 20)$$

$$= 85 - (15 + 35 + 20) = 15.$$

453. Number of students who took any one or both the languages

